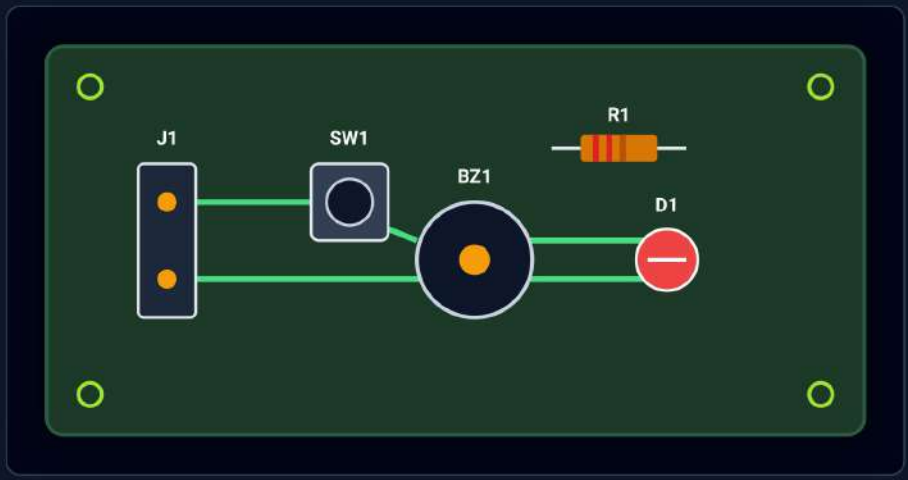
KICAD PCB DESIGN PROJECT

EMERGENCY PANIC ALARM

Schematic | PCB Layout | 3D View | Gerber Output



Rajvardhan Singh
ECE | VLSI Specialist

HARDWARE DESIGN	PROJECT TYPE	DOCUMENT STATUS
KiCad v8.0	Emergency Panic Alarm	Complete

PROJECT OVERVIEW

The Emergency Panic Alarm is a robust 5V alert circuit engineered for immediate visual and acoustic warning signaling. Designed for reliability and swift activation, pressing the momentary switch instantly triggers the high-decibel piezo buzzer and high-luminosity warning LED.

INPUT STAGE
5V INPUT
DC Power connector via J1 header interface

TRIGGER STAGE
PUSH BUTTON
Momentary normally-open tactile switch SW1

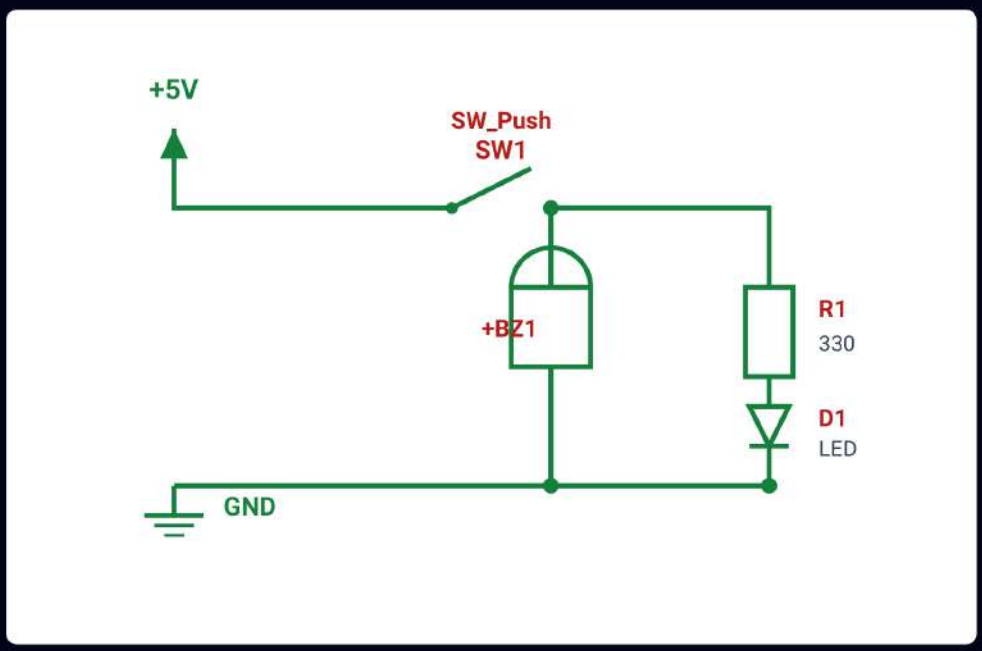
OUTPUT STAGE
ALERT ACTIVE
Simultaneous Buzzer audio & LED indicator

BILL OF MATERIALS (BOM)

Reference	Description / Specification	Qty	Footprint Type
J1	2-pin 5V DC Power Input Connector	1	PinHeader_1x02_P2.54mm
SW1	Normally-Open Push Button Switch	1	SW_PUSH_6mm
BZ1	5V Active Piezo Acoustic Buzzer	1	Buzzer_12x9.5mm_P7.6mm
D1	5mm High-Brightness Red LED	1	LED_D5.0mm
R1	330 Ω Current Limiting Resistor (1/4W)	1	R_Axial_DIN0207_L6.3mm_D2.5mm

CIRCUIT SCHEMATIC

Complete circuit schematic illustrating power distribution flags, switch activation logic, and parallel alert component driving.

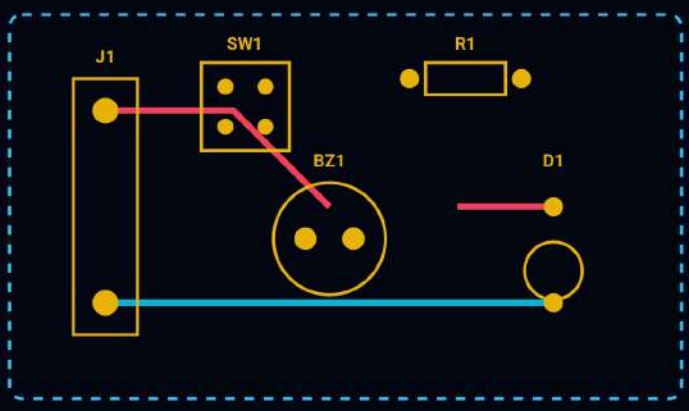


PWR_FLAG Verification: Passed

ERC Status: 0 Errors / 0 Warnings

PCB LAYOUT & ROUTING

2-layer printed circuit board design with optimized copper traces, robust ground plane pours, and zero unrouted connections.



MANUFACTURING OUTPUT (GERBER FILES)

Gerber Output File	Layer / Purpose
Emergency_Panic_Alarm-F_Cu.gbr	Top Copper Layer
Emergency_Panic_Alarm-B_Cu.gbr	Bottom Ground Plane
Emergency_Panic_Alarm-F_Mask.gbr	Top Solder Mask
Emergency_Panic_Alarm-B_Mask.gbr	Bottom Solder Mask
Emergency_Panic_Alarm-F_Silks.gbr	Top Silkscreen
Emergency_Panic_Alarm-Edge_Cuts.gbr	Board Outline
Emergency_Panic_Alarm-PTH.drl	Plated Drill Holes
Emergency_Panic_Alarm-NPTH.drl	Non-Plated Mounting Holes

VERIFIED & COMPLETE

PROJECT COMPLETE

Emergency Panic Alarm PCB Layout & Schematic Verification Successfully Executed.

DESIGNED & DEVELOPED BY

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Designed in KiCad | Ready for Fabrication